

Lab4: Insight into PHP and SQL

Objective

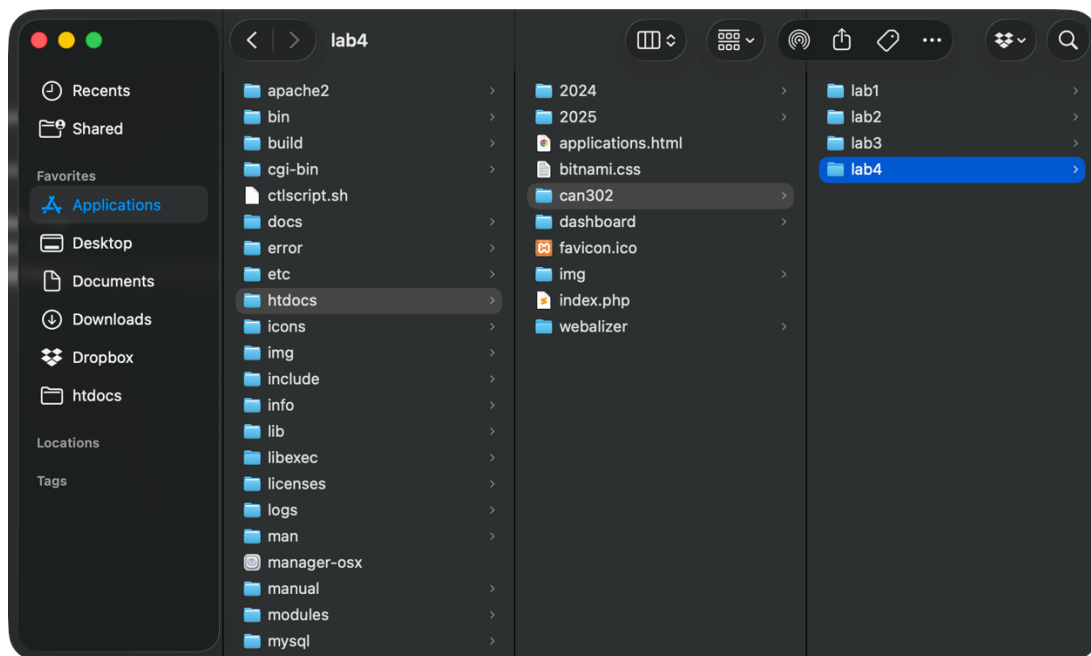
- Demonstrate the basic working knowledge of PHP
- Utilize SQL to create, query, and manage databases
- Integrate SQL with PHP to build dynamic web pages

Tips

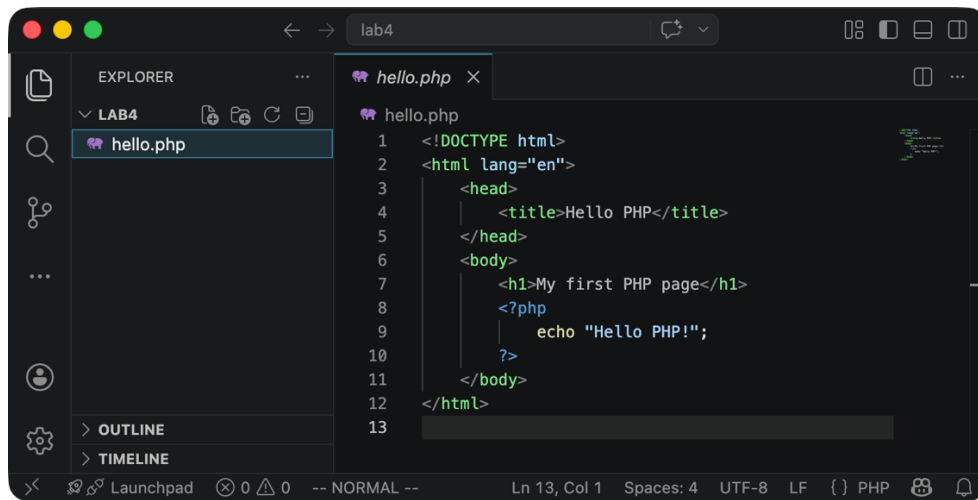
- These labs are expected to take more than the allocated 2 hours. You should try the tasks before the lab. Practice makes code perfect.
- Before the lab, you could post your questions in LMO forum, and you are welcome to offer solutions to the questions in LMO.
- During the lab, you could discuss with TAs for your questions. Face to face discussion will have a relatively high efficiency.

A. PHP Basics

1. Create a new subfolder "lab4" under folder "can 302" (The following screenshot is in macOS, but same operations can be applied to Windows):

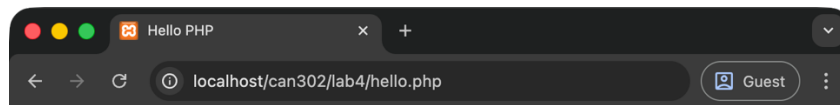


2. Use VS Code to open "lab4" and create a new file "hello.php" as follows:



```
1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <title>Hello PHP</title>
5   </head>
6   <body>
7     <h1>My first PHP page</h1>
8     <?php
9       echo "Hello PHP!";
10    ?>
11   </body>
12 </html>
13
```

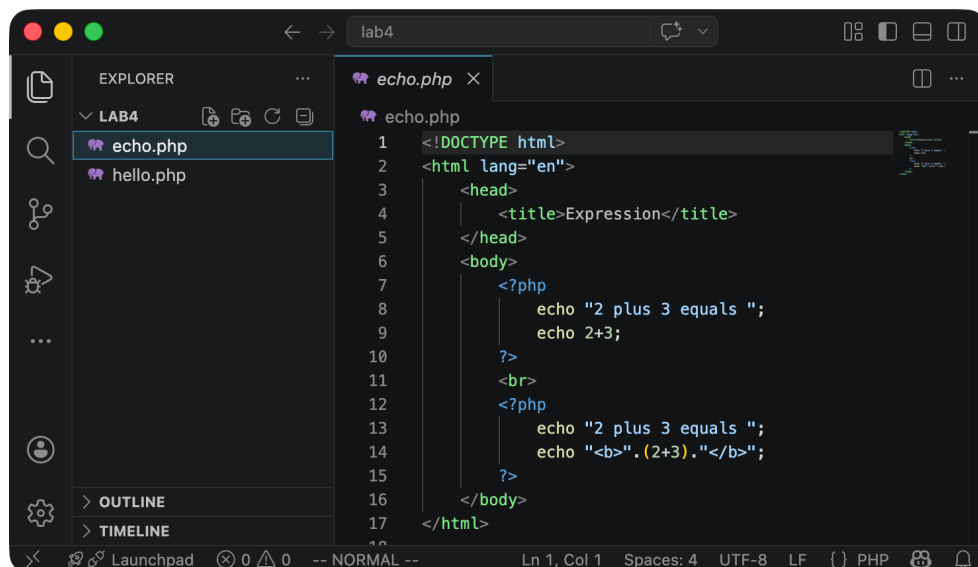
3. Access the webpage in your browser via URL: <http://localhost/can302/lab4/hello.php>



My first PHP page

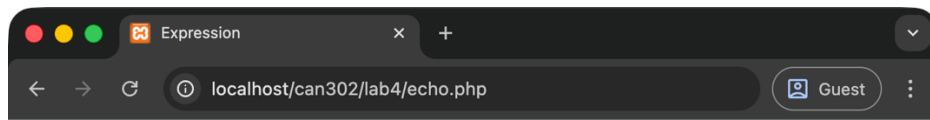
Hello PHP!

4. Create a new file "echo.php" under "lab4" as follows:

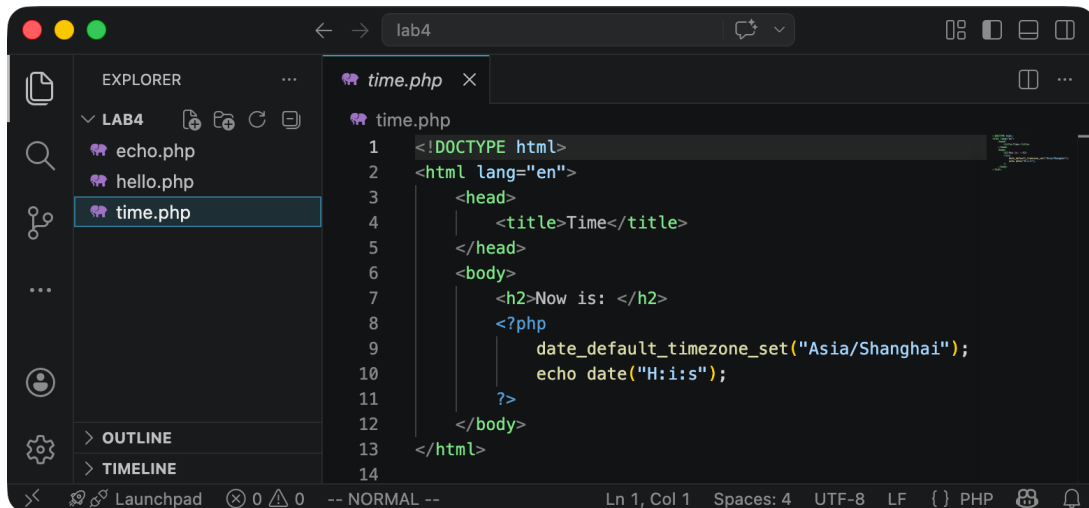


```
1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <title>Expression</title>
5   </head>
6   <body>
7     <?php
8       echo "2 plus 3 equals ";
9       echo 2+3;
10    ?>
11     <br>
12     <?php
13       echo "2 plus 3 equals ";
14       echo "<b>".(2+3)."</b>";
15     ?>
16   </body>
17 </html>
18
```

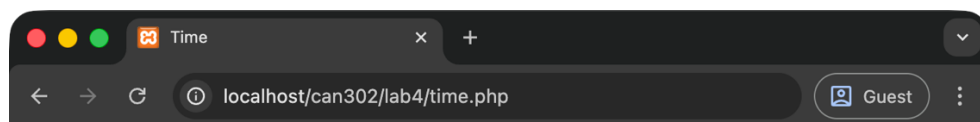
5. Access the webpage in your browser via URL: <http://localhost/can302/lab4/echo.php>



6. Create a new file "time.php" under "lab4" as follows:



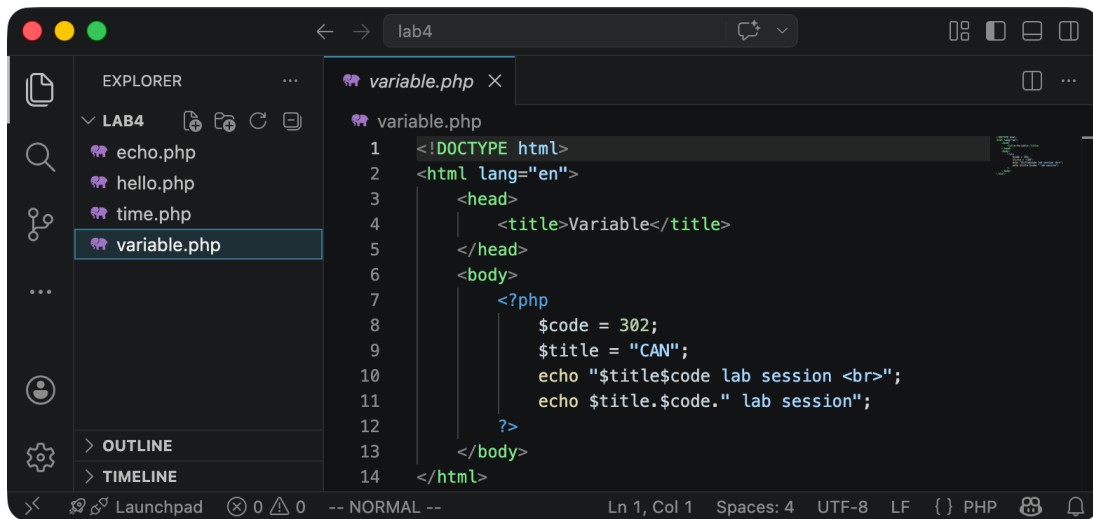
7. Access the webpage in your browser via URL: <http://localhost/can302/lab4/time.php>



Now is:

15:17:12

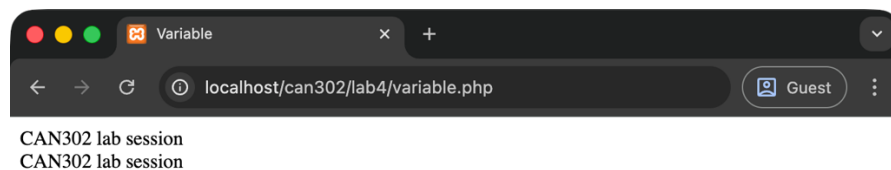
8. Create a new file "variable.php" under "lab4" as follows:



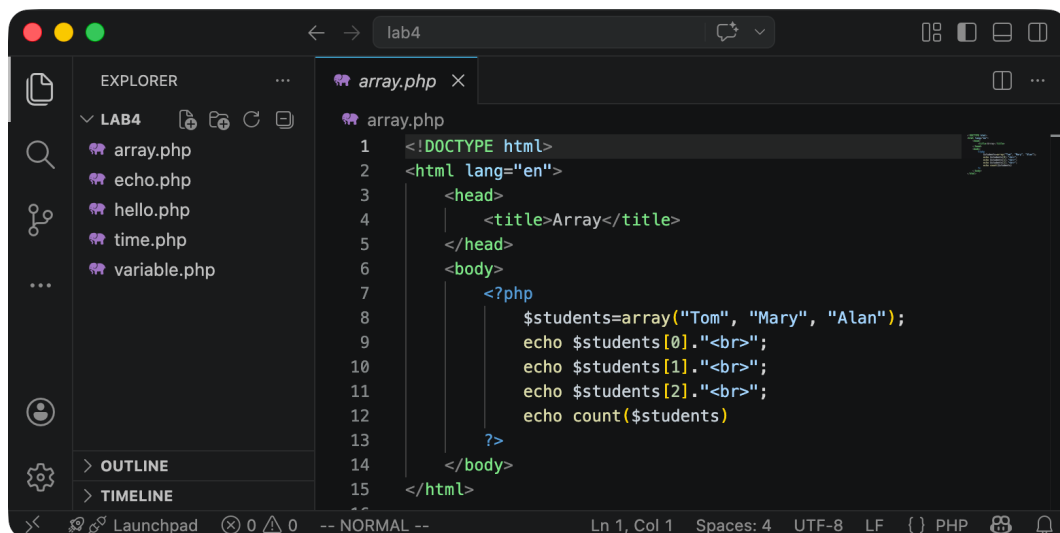
The screenshot shows a code editor with a file explorer on the left. The file explorer shows a directory named 'LAB4' containing files: 'echo.php', 'hello.php', 'time.php', and 'variable.php'. The 'variable.php' file is selected. The main editor area shows the following PHP code:

```
1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <title>Variable</title>
5   </head>
6   <body>
7     <?php
8       $code = 302;
9       $title = "CAN";
10      echo "$title$code lab session <br>";
11      echo $title.$code." lab session";
12    ?>
13  </body>
14 </html>
```

9. Access the webpage in your browser via URL: <http://localhost/can302/lab4/variable.php>



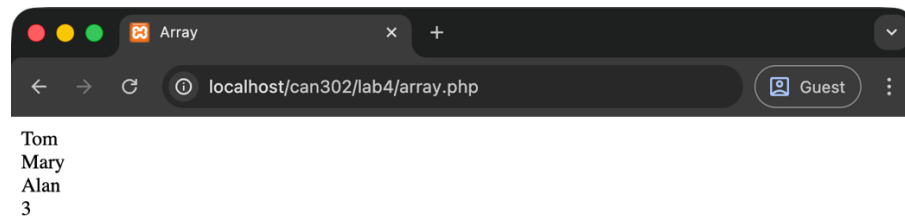
10. Create a new file "array.php" under "lab4" as follows:



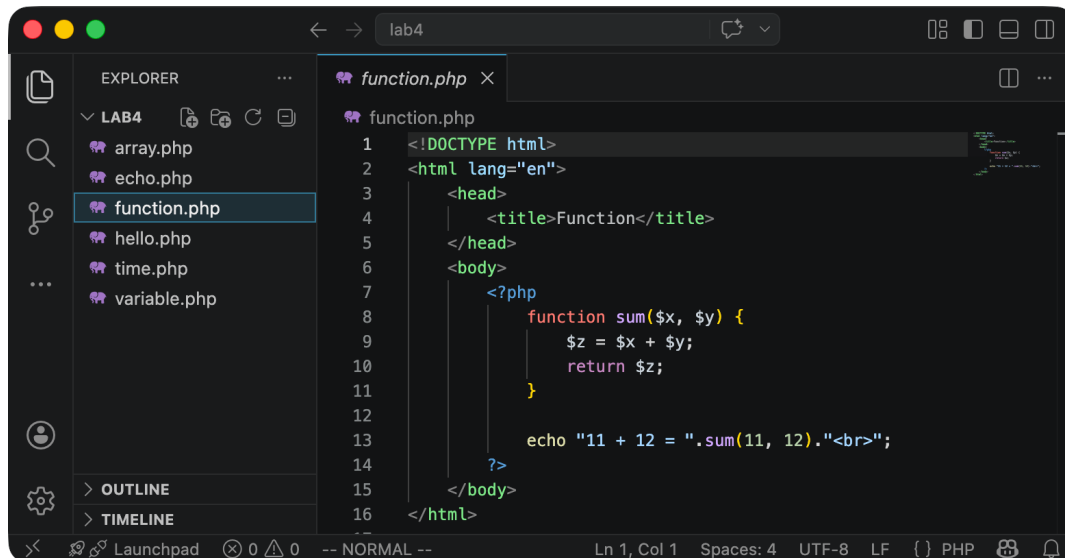
The screenshot shows a code editor with a file explorer on the left. The file explorer shows a directory named 'LAB4' containing files: 'array.php', 'echo.php', 'hello.php', 'time.php', and 'variable.php'. The 'array.php' file is selected. The main editor area shows the following PHP code:

```
1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <title>Array</title>
5   </head>
6   <body>
7     <?php
8       $students=array("Tom", "Mary", "Alan");
9       echo $students[0]."<br>";
10      echo $students[1]."<br>";
11      echo $students[2]."<br>";
12      echo count($students)
13    ?>
14  </body>
15 </html>
```

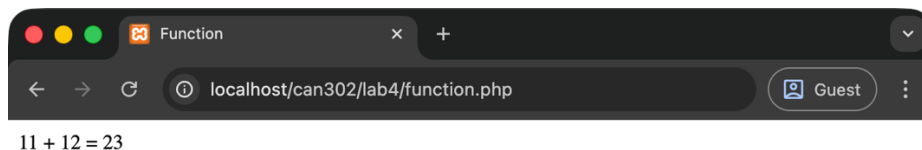

11. Access the webpage in your browser via URL: <http://localhost/can302/lab4/array.php>



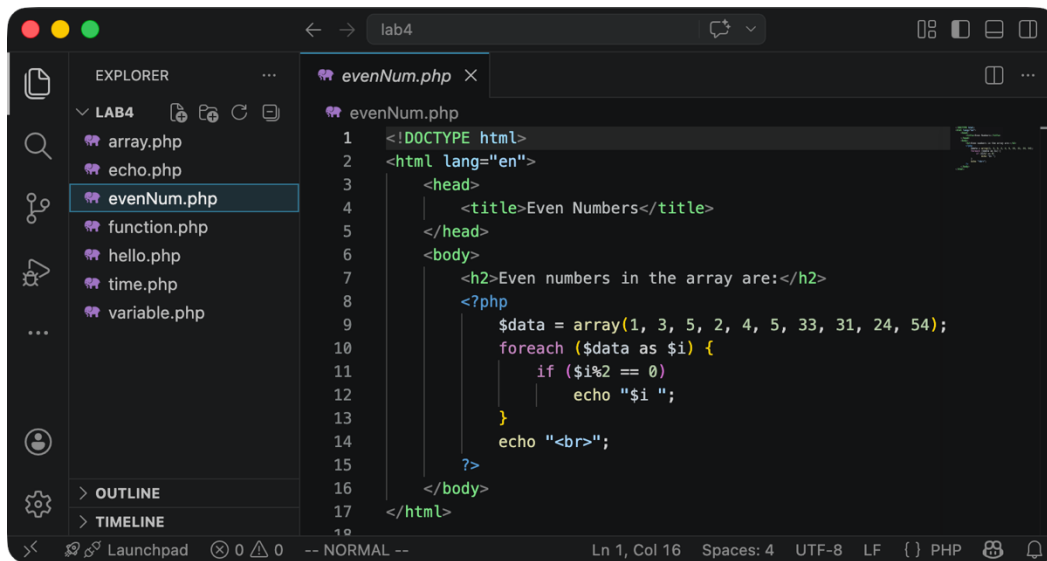
12. Create a new file "function.php" under "lab4" as follows:



13. Access the webpage in your browser via URL: <http://localhost/can302/lab4/function.php>



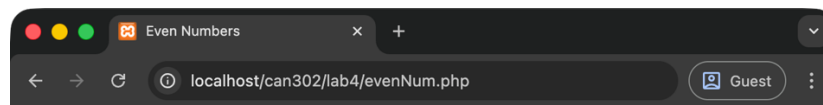
14. Create a new file "evenNum.php" under "lab4" as follows:



The screenshot shows a code editor with a file explorer on the left. The file explorer shows a directory named 'LAB4' containing several PHP files: array.php, echo.php, evenNum.php (selected), function.php, hello.php, time.php, and variable.php. The main editor window shows the content of evenNum.php, which is an HTML document with a title 'Even Numbers' and a body containing a heading 'Even numbers in the array are:' followed by a PHP script that iterates over an array of numbers and echoes the even ones.

```
1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <title>Even Numbers</title>
5   </head>
6   <body>
7     <h2>Even numbers in the array are:</h2>
8     <?php
9       $data = array(1, 3, 5, 2, 4, 5, 33, 31, 24, 54);
10      foreach ($data as $i) {
11        if ($i%2 == 0)
12          echo "$i ";
13      }
14      echo "<br>";
15    <?>
16  </body>
17 </html>
```

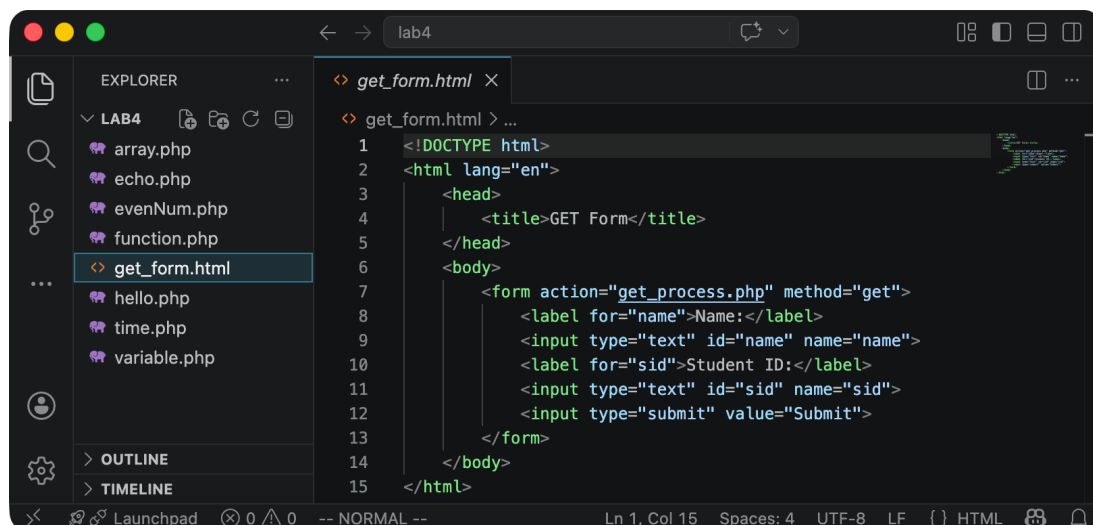
15. Access the webpage in your browser via URL: <http://localhost/can302/lab4/evenNum.php>



Even numbers in the array are:

2 4 24 54

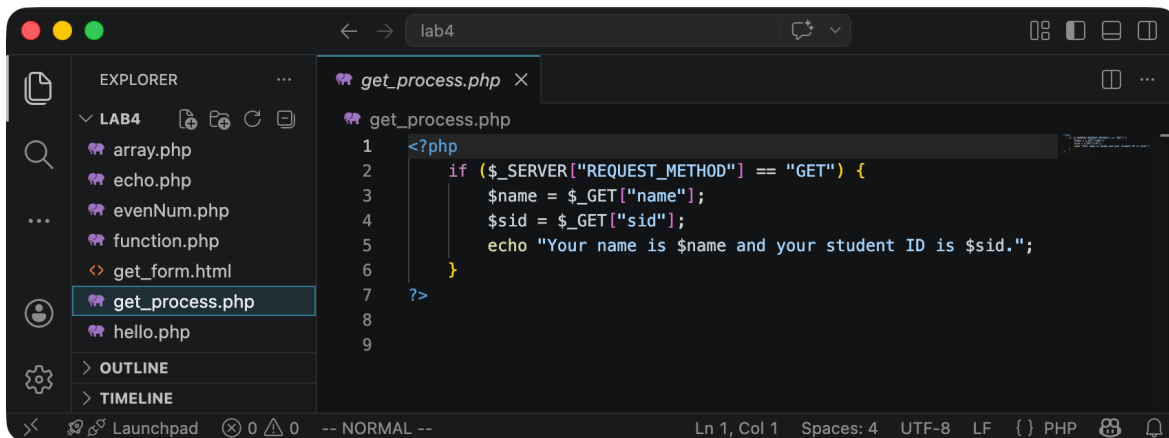
16. Create a new file "get_form.html" under "lab4" as follows:



The screenshot shows a code editor with a file explorer on the left. The file explorer shows a directory named 'LAB4' containing several PHP files: array.php, echo.php, evenNum.php, function.php, get_form.html (selected), hello.php, time.php, and variable.php. The main editor window shows the content of get_form.html, which is an HTML document with a title 'GET Form' and a body containing a form with two text inputs and a submit button.

```
1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <title>GET Form</title>
5   </head>
6   <body>
7     <form action="get_process.php" method="get">
8       <label for="name">Name:</label>
9       <input type="text" id="name" name="name">
10      <label for="sid">Student ID:</label>
11      <input type="text" id="sid" name="sid">
12      <input type="submit" value="Submit">
13    </form>
14  </body>
15 </html>
```

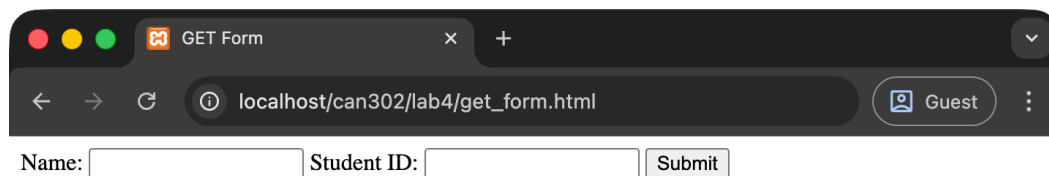
17. Create a new file "get_process.php" under "lab4" as follows:



The screenshot shows a code editor with a file explorer on the left. The file explorer shows a directory named 'LAB4' containing several files: array.php, echo.php, evenNum.php, function.php, get_form.html, get_process.php (selected), and hello.php. The main editor window shows the content of 'get_process.php' with the following PHP code:

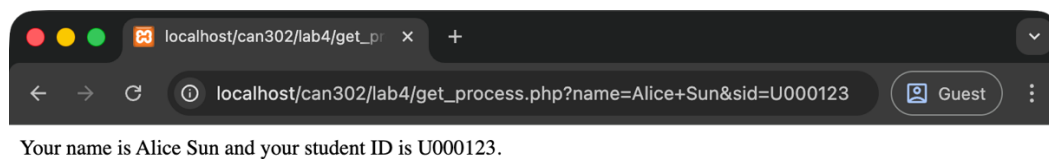
```
1 <?php
2     if ($_SERVER["REQUEST_METHOD"] == "GET") {
3         $name = $_GET["name"];
4         $sid = $_GET["sid"];
5         echo "Your name is $name and your student ID is $sid.";
6     }
7 }
8
9 ?>
```

18. Access the webpage in your browser via URL: http://localhost/can302/lab4/get_form.html



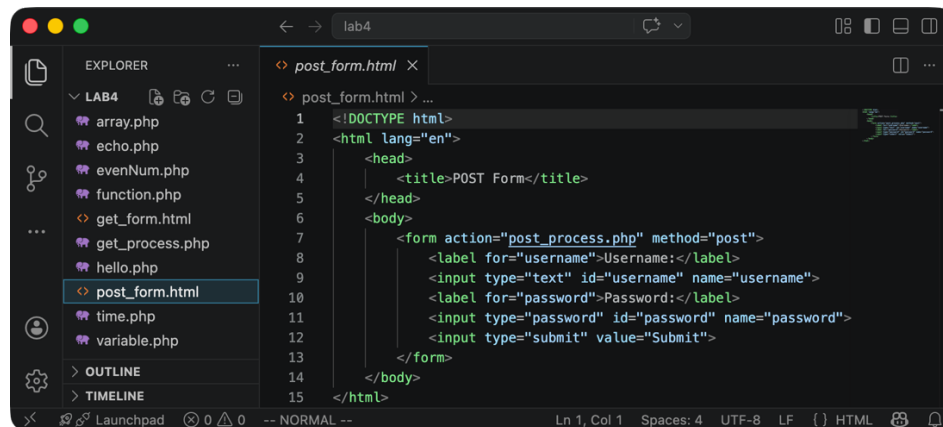
The screenshot shows a web browser window with the title 'GET Form'. The address bar shows the URL 'localhost/can302/lab4/get_form.html'. The page content includes a form with two input fields: 'Name:' and 'Student ID:', followed by a 'Submit' button.

19. Input a name and a student ID (i.e., Alice Sun and U000123), and click the "Submit" button:



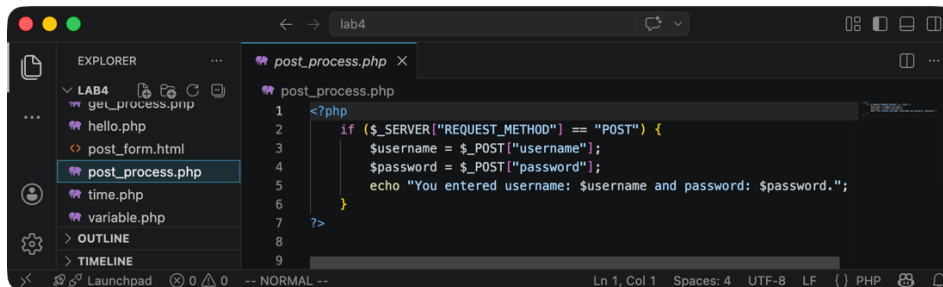
The screenshot shows a web browser window with the title 'localhost/can302/lab4/get_pr'. The address bar shows the URL 'localhost/can302/lab4/get_process.php?name=Alice+Sun&sid=U000123'. The page content displays the output of the PHP script: 'Your name is Alice Sun and your student ID is U000123.'

20. Create a new file "post_form.html" under "lab4" as follows:



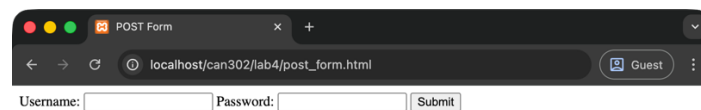
```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>POST Form</title>
</head>
<body>
  <form action="post_process.php" method="post">
    <label for="username">Username:</label>
    <input type="text" id="username" name="username">
    <label for="password">Password:</label>
    <input type="password" id="password" name="password">
    <input type="submit" value="Submit">
  </form>
</body>
</html>
```

21. Create a new file "post_process.php" under "lab4" as follows:

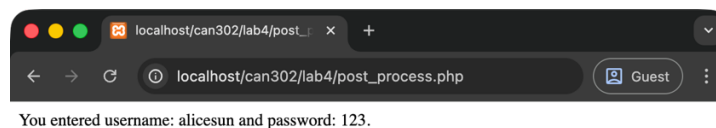


```
<?php
if ($_SERVER["REQUEST_METHOD"] == "POST") {
  $username = $_POST["username"];
  $password = $_POST["password"];
  echo "You entered username: $username and password: $password.";
}
```

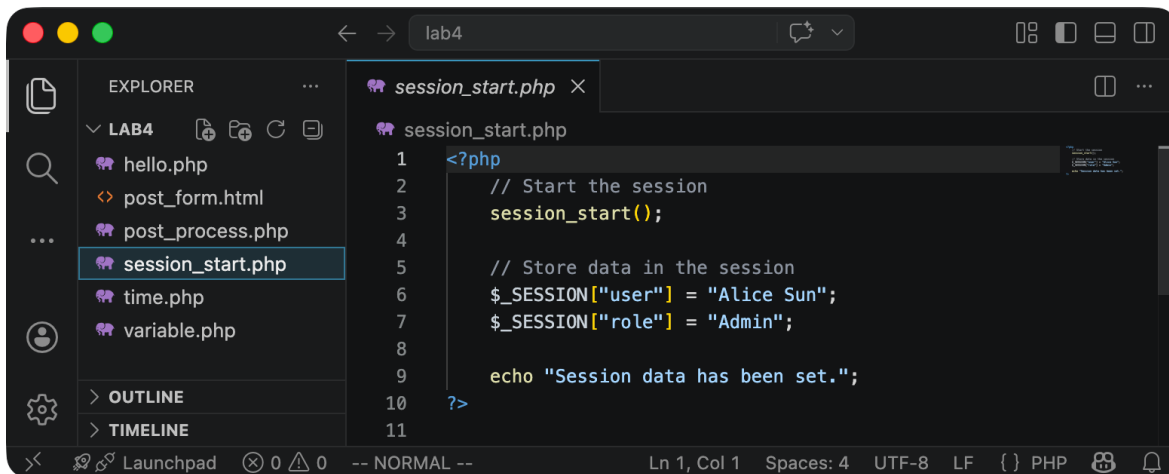
22. Access the webpage in your browser via URL: http://localhost/can302/lab4/post_form.html



23. Input a username and a password (i.e., alicesun and 123), and click the "Submit" button:



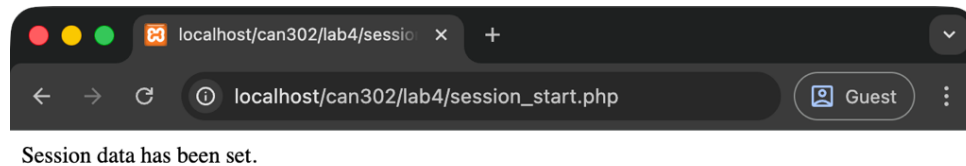
24. Create a new file "session_start.php" under "lab4" as follows:



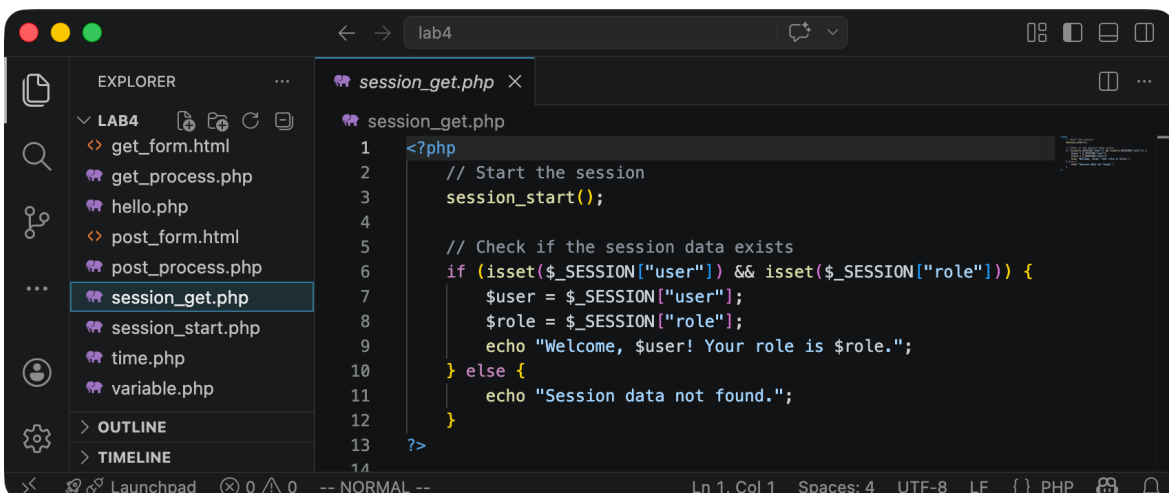
The screenshot shows a code editor with a file explorer on the left. The file explorer shows a directory named "LAB4" containing several files: hello.php, post_form.html, post_process.php, session_start.php (highlighted), time.php, and variable.php. The main editor area shows the content of session_start.php:

```
1 <?php
2 // Start the session
3 session_start();
4
5 // Store data in the session
6 $_SESSION["user"] = "Alice Sun";
7 $_SESSION["role"] = "Admin";
8
9 echo "Session data has been set.";
10 ?>
11
```

25. Access the webpage in your browser via URL: http://localhost/can302/lab4/session_start.php



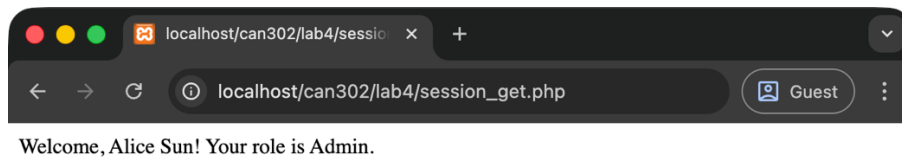
26. Create a new file "session_get.php" under "lab4" as follows:



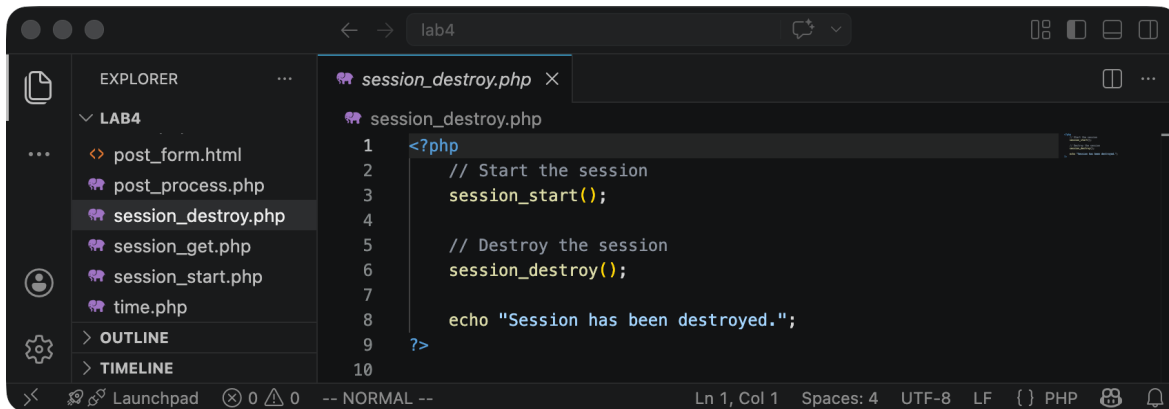
The screenshot shows a code editor with a file explorer on the left. The file explorer shows a directory named "LAB4" containing several files: get_form.html, get_process.php, hello.php, post_form.html, post_process.php, session_get.php (highlighted), session_start.php, time.php, and variable.php. The main editor area shows the content of session_get.php:

```
1 <?php
2 // Start the session
3 session_start();
4
5 // Check if the session data exists
6 if (isset($_SESSION["user"]) && isset($_SESSION["role"])) {
7     $user = $_SESSION["user"];
8     $role = $_SESSION["role"];
9     echo "Welcome, $user! Your role is $role.";
10 } else {
11     echo "Session data not found.";
12 }
13 ?>
14
```

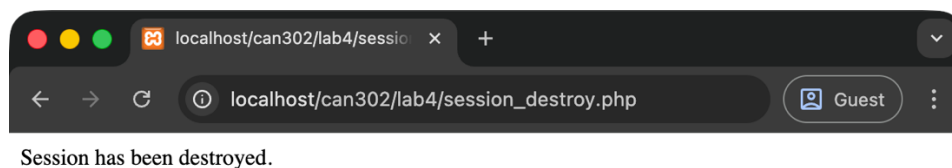
27. Access the webpage in your browser via URL: http://localhost/can302/lab4/session_get.php



28. Create a new file "session_destroy.php" under "lab4" as follows:

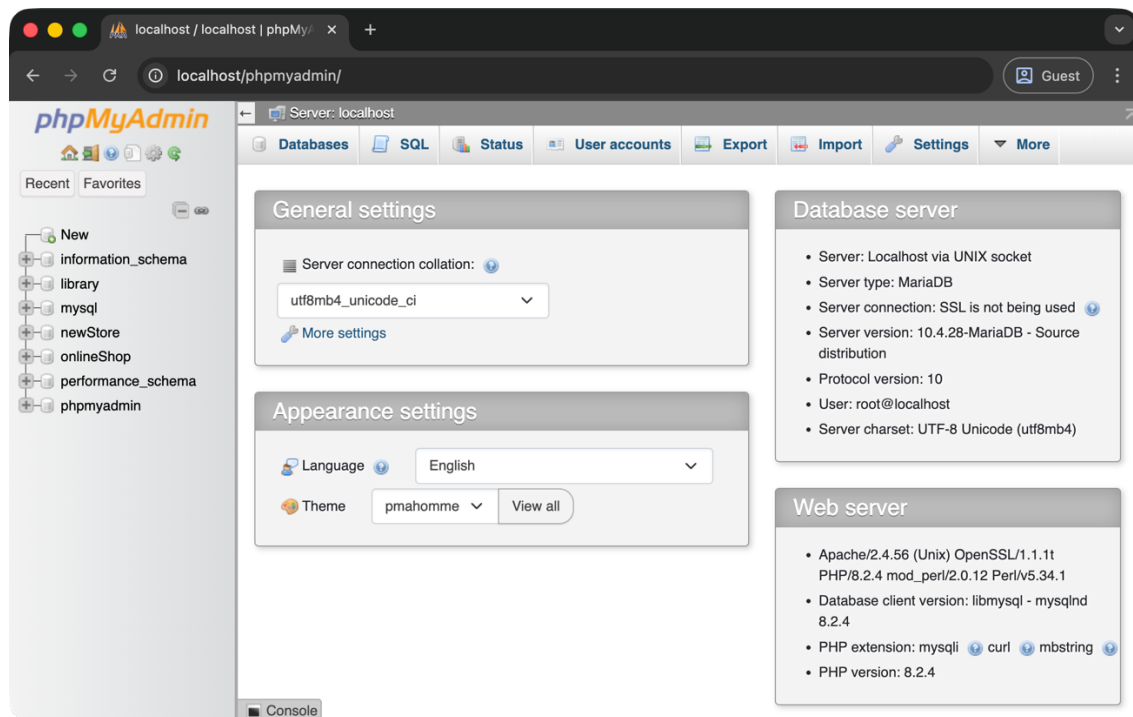


29. Access the webpage in your browser via URL: http://localhost/can302/lab4/session_destroy.php

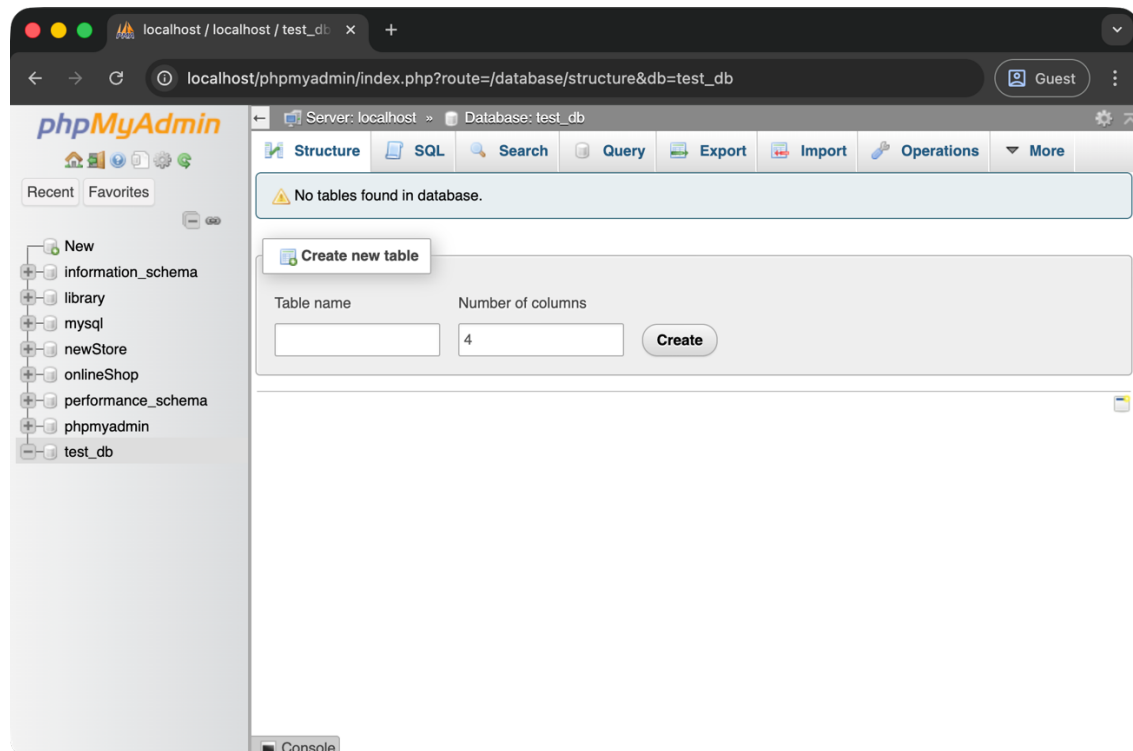


B. SQL Basics

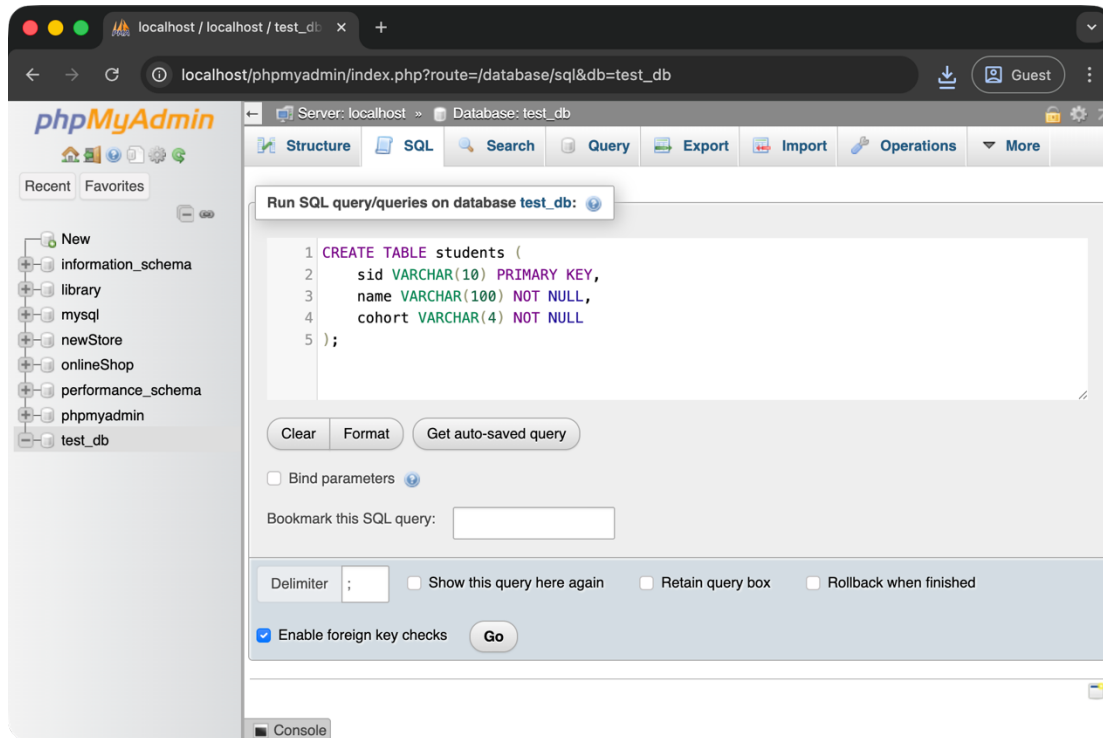
1. Open the browser and visit <http://localhost/phpmyadmin>



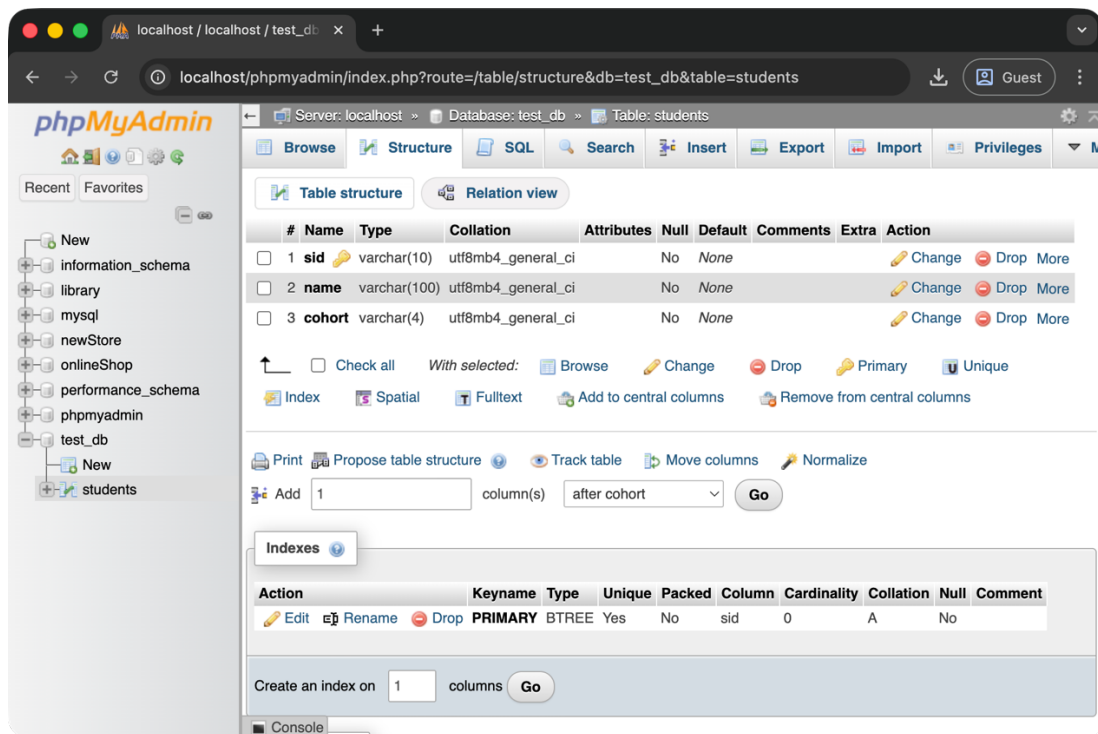
2. In the phpMyAdmin interface, click "New" in the left panel, enter "test_db" in the "Database" input box, and then click the "Create" button. Then, the "test_db" will appear in the left panel:



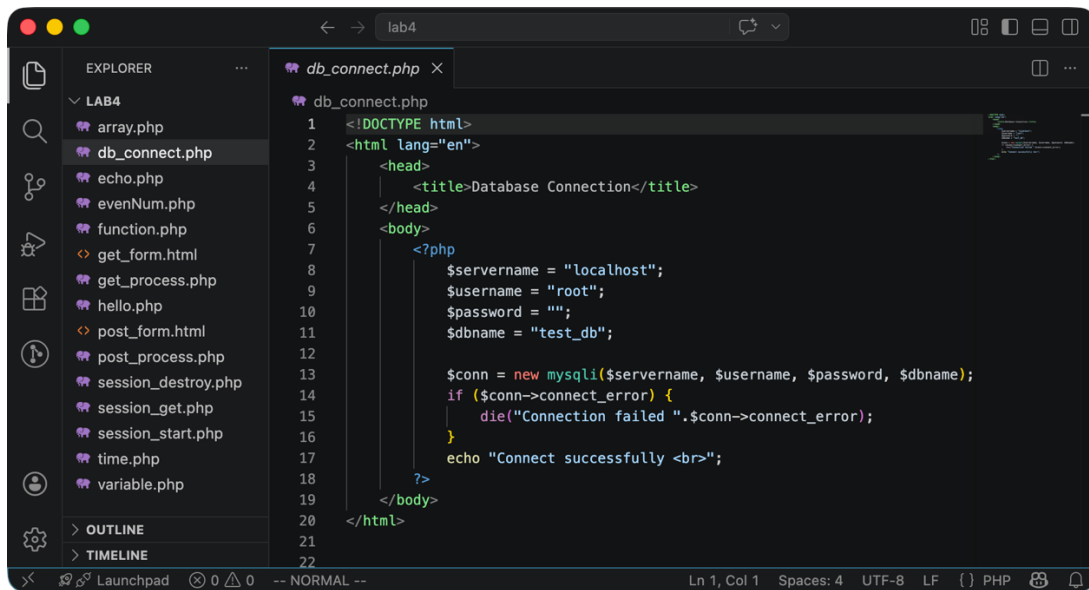
3. Select the newly created "test_db" database and click the "SQL" tab. Then, enter the following SQL statement in the text box, and click the "Go" button:



4. The statement creates a table named "students" with three fields: sid, name, and cohort:



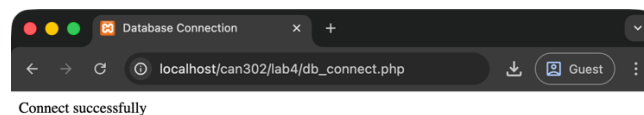
5. Create a new file "db_connect.php" under "lab4" as follows:



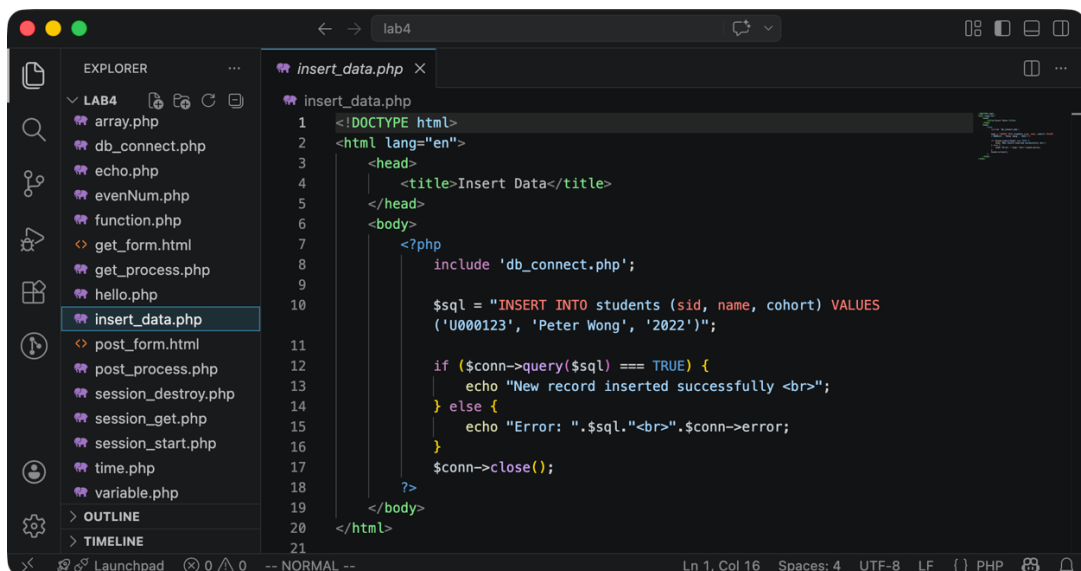
The screenshot shows a code editor with a file explorer on the left and a code editor on the right. The file explorer shows a directory named "LAB4" containing several files, including "db_connect.php". The code editor shows the contents of "db_connect.php", which is an HTML document with a title "Database Connection" and a PHP script that connects to a MySQL database. The PHP script uses the mysqli extension to connect to a database named "test_db" on localhost with the username "root" and an empty password. If the connection fails, it displays an error message. If successful, it displays "Connect successfully".

```
1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <title>Database Connection</title>
5   </head>
6   <body>
7     <?php
8       $servername = "localhost";
9       $username = "root";
10      $password = "";
11      $dbname = "test_db";
12
13      $conn = new mysqli($servername, $username, $password, $dbname);
14      if ($conn->connect_error) {
15        die("Connection failed ". $conn->connect_error);
16      }
17      echo "Connect successfully <br>";
18    <?>
19  </body>
20 </html>
```

6. Open the browser and visit http://localhost/can302/lab4/db_connect.php



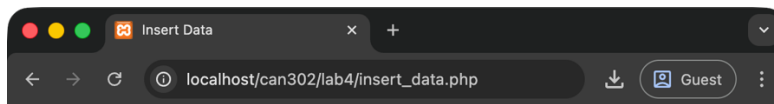
7. Create a new file "insert_data.php" under "lab4" as follows:



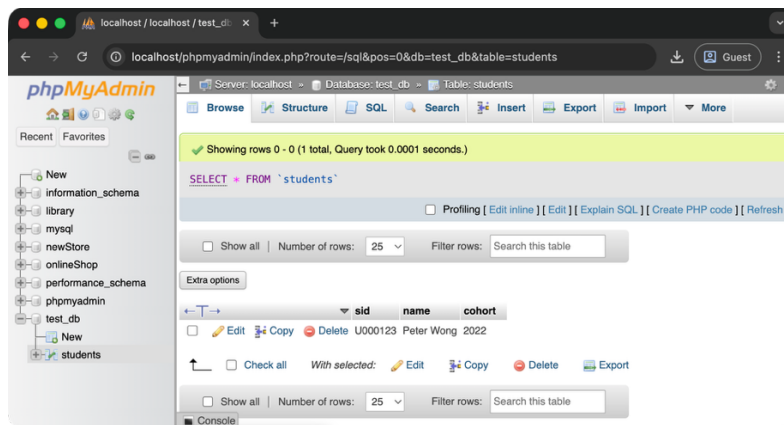
The screenshot shows a code editor with a file explorer on the left and a code editor on the right. The file explorer shows a directory named "LAB4" containing several files, including "insert_data.php". The code editor shows the contents of "insert_data.php", which is an HTML document with a title "Insert Data" and a PHP script that inserts data into a MySQL database. The PHP script includes "db_connect.php" and uses the mysqli extension to insert a new record into the "students" table. If the insertion is successful, it displays "New record inserted successfully". If there is an error, it displays the error message.

```
1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <title>Insert Data</title>
5   </head>
6   <body>
7     <?php
8       include 'db_connect.php';
9
10      $sql = "INSERT INTO students (sid, name, cohort) VALUES
11      ('U000123', 'Peter Wong', '2022')";
12
13      if ($conn->query($sql) === TRUE) {
14        echo "New record inserted successfully <br>";
15      } else {
16        echo "Error: ". $sql. "<br>". $conn->error;
17      }
18      $conn->close();
19    <?>
20  </body>
21 </html>
```

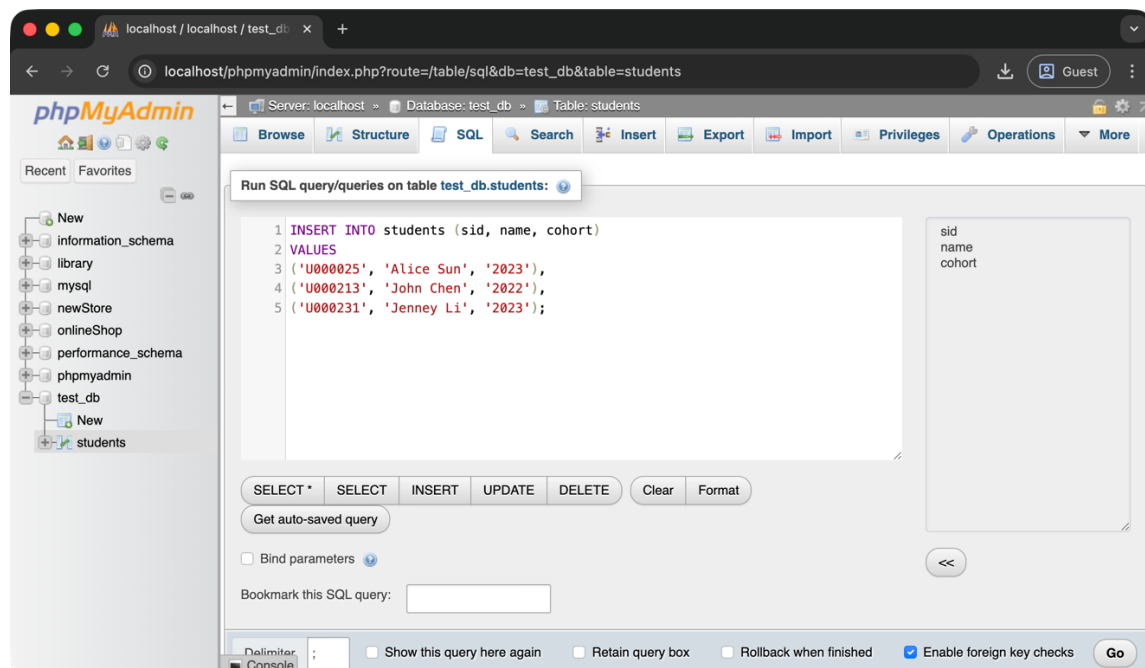
8. Open the browser and visit http://localhost/can302/lab4/insert_data.php



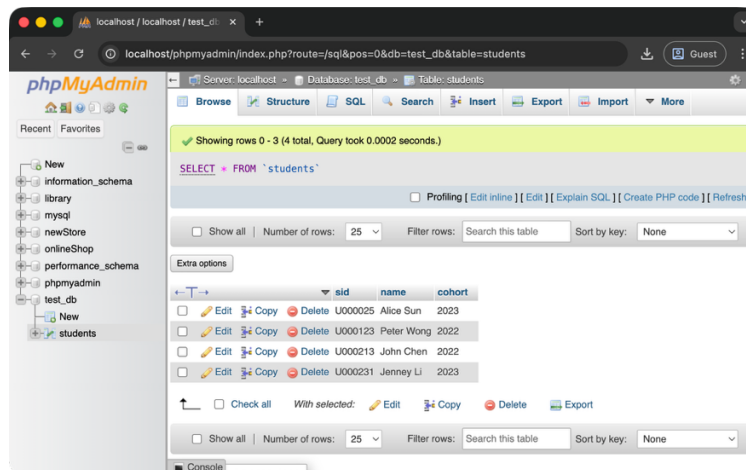
9. Visit <http://localhost/phpmyadmin>, and check the newly inserted record:



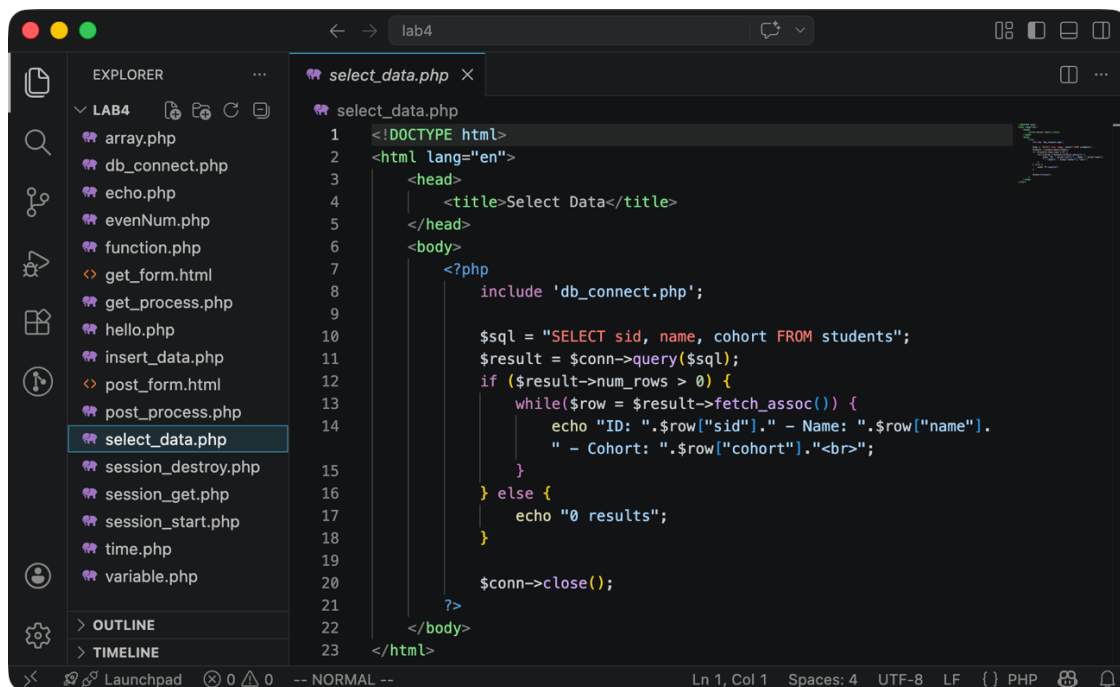
10. Insert more records in the table "students" as follows, and click the "Go" button:



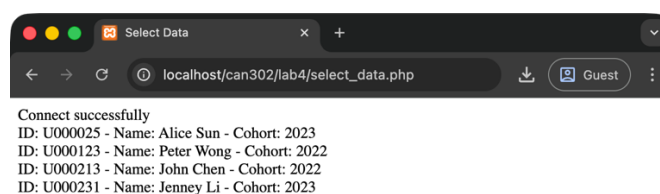
11. Check the newly inserted records:



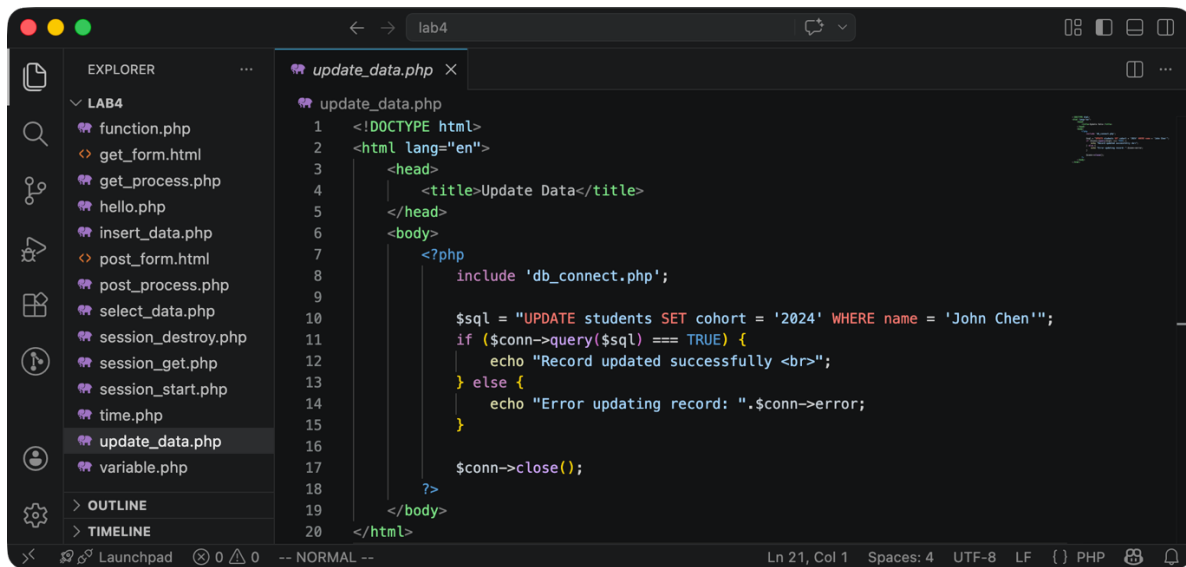
12. Create a new file "select_data.php" under "lab4" as follows:



13. Open the browser and visit http://localhost/can302/lab4/select_data.php

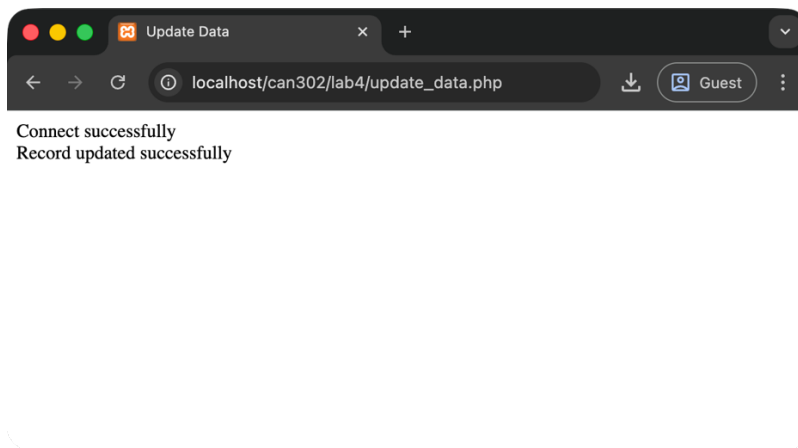


14. Create a new file "update_data.php" under "lab4" as follows:

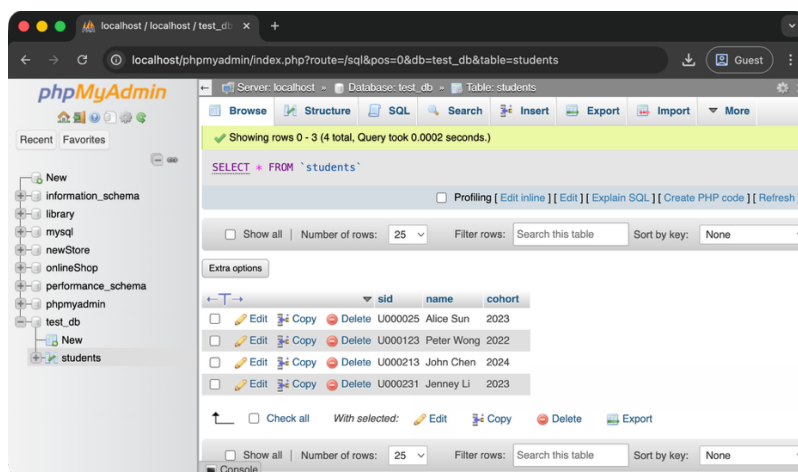


```
1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <title>Update Data</title>
5   </head>
6   <body>
7     <?php
8       include 'db_connect.php';
9
10      $sql = "UPDATE students SET cohort = '2024' WHERE name = 'John Chen'";
11      if ($conn->query($sql) === TRUE) {
12        echo "Record updated successfully <br>";
13      } else {
14        echo "Error updating record: " . $conn->error;
15      }
16
17      $conn->close();
18    ?>
19   </body>
20 </html>
```

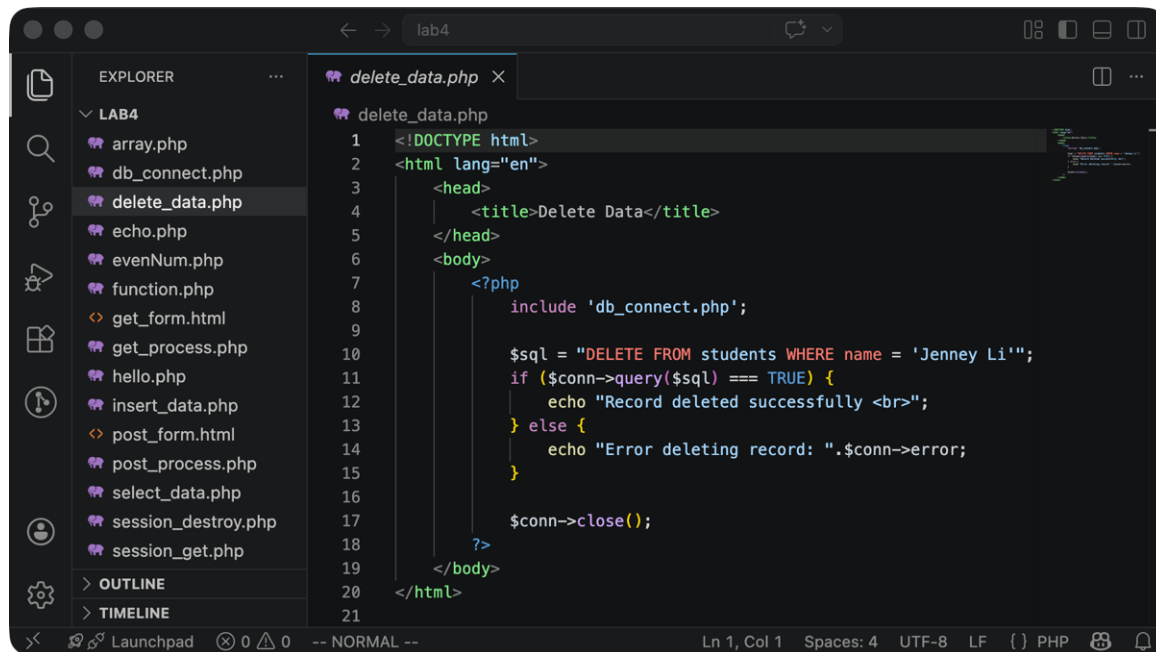
15. Open the browser and visit http://localhost/can302/lab4/update_data.php



16. Visit <http://localhost/phpmyadmin>, and check the updated record:

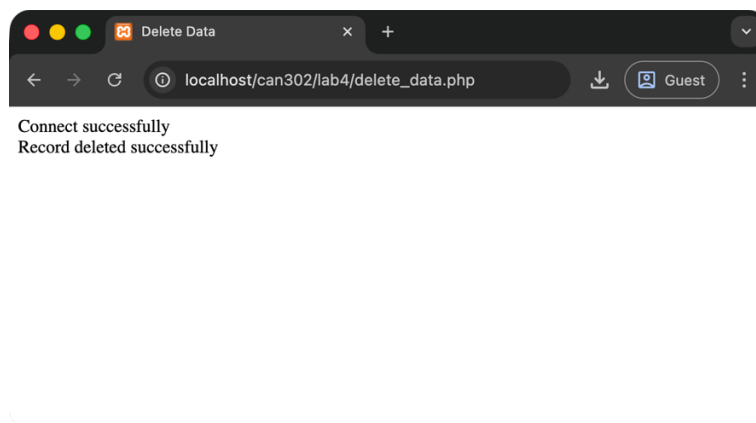


17. Create a new file "delete_data.php" under "lab4" as follows:

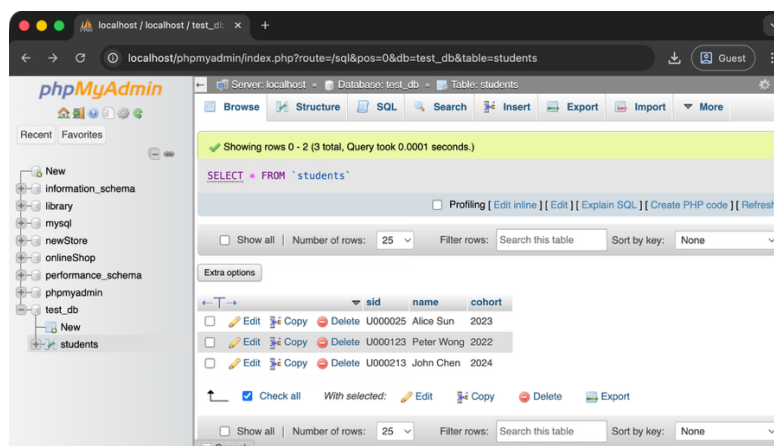


```
1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <title>Delete Data</title>
5   </head>
6   <body>
7     <?php
8       include 'db_connect.php';
9
10      $sql = "DELETE FROM students WHERE name = 'Jenney Li'";
11      if ($conn->query($sql) === TRUE) {
12        echo "Record deleted successfully <br>";
13      } else {
14        echo "Error deleting record: ".$conn->error;
15      }
16
17      $conn->close();
18    ?>
19  </body>
20 </html>
21
```

18. Open the browser and visit http://localhost/can302/lab4/delete_data.php

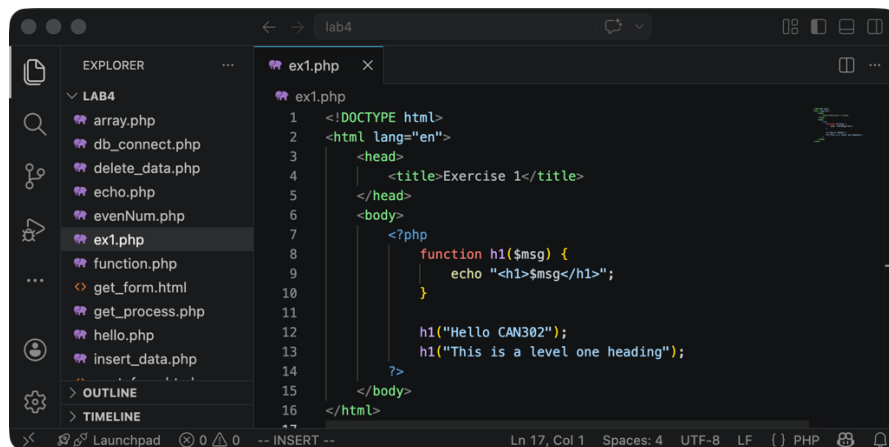


19. Visit <http://localhost/phpmyadmin>, and check if the record is deleted:

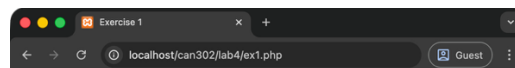


C. Hand-on Exercises

1. In this task, you are going to use function to simplify web page generation:
 - (a) Create a new file "ex1.php" under "lab4";
 - (b) Declare a function "h1" which accepts a string as argument and print out the string in HTML level 1 heading, i.e., <h1>...</h1>.



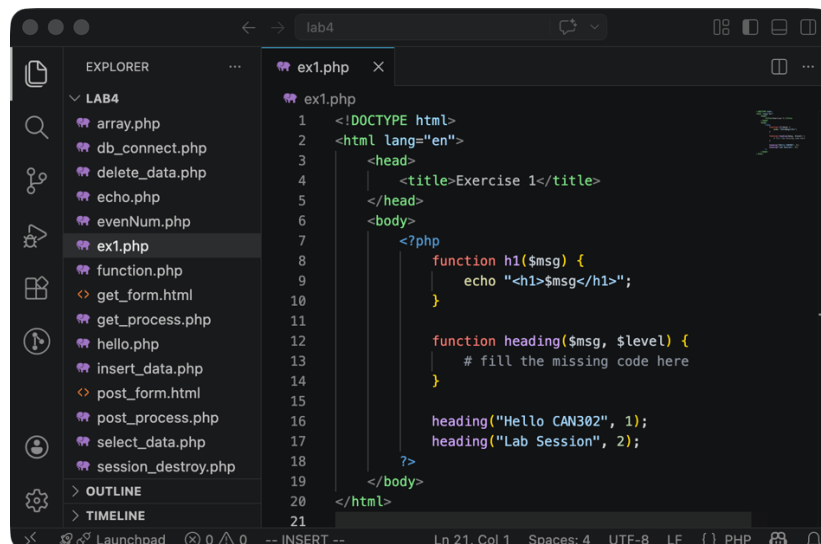
```
1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <title>Exercise 1</title>
5   </head>
6   <body>
7     <?php
8       function h1($msg) {
9         echo "<h1>$msg</h1>";
10      }
11
12      h1("Hello CAN302");
13      h1("This is a level one heading");
14    ?>
15  </body>
16 </html>
```



Hello CAN302

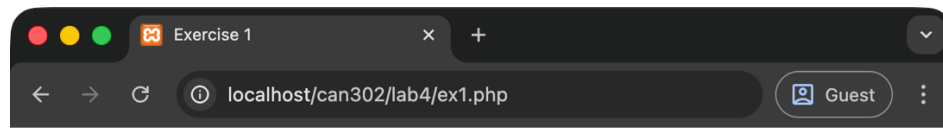
This is a level one heading

- (c) Declare another function call "heading" which accepts two arguments, a message and a heading level (you may assume the highest heading level is 6):



```
1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <title>Exercise 1</title>
5   </head>
6   <body>
7     <?php
8       function h1($msg) {
9         echo "<h1>$msg</h1>";
10      }
11
12       function heading($msg, $level) {
13         # fill the missing code here
14      }
15
16       heading("Hello CAN302", 1);
17       heading("Lab Session", 2);
18    ?>
19  </body>
20 </html>
21
```

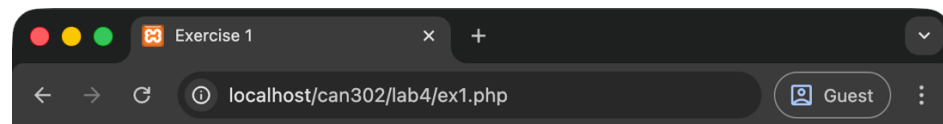
And it should look like:



Hello CAN302

Lab Session

(d) Try to generate the following page with for-loop and the function "heading":



Hello CAN302

Hello CAN302

Hello CAN302

Hello CAN302

Hello CAN302

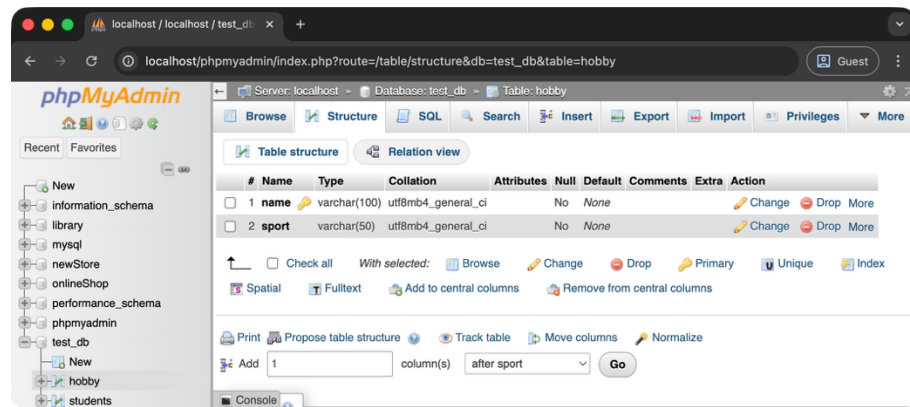
Hello CAN302

2. Practice database connection and table creation.

(a) Recall that we have created a database "test_db" and a table "students" with 3 columns:

sid	name	cohort
U000025	Alice Sun	2023
U000123	Peter Wong	2022
U000213	John Chen	2024

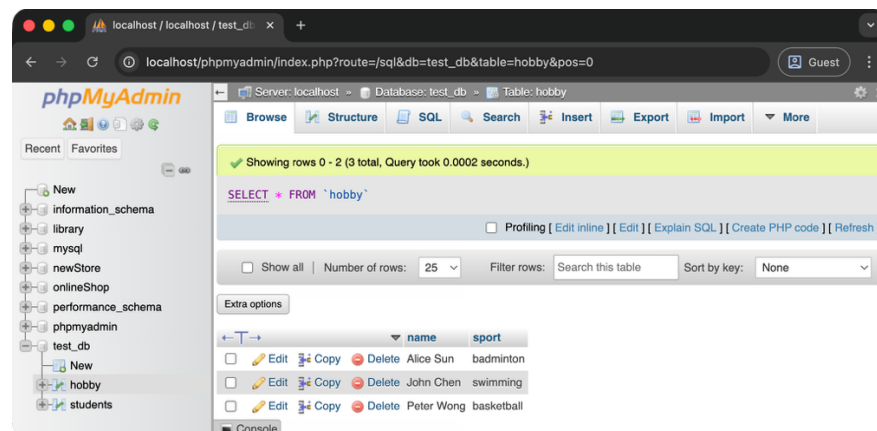
(b) Create a PHP script named "ex2_1.php" that connects to the "test_db" database and creates a table named "hobby" with columns "name" (VARCHAR (100) , primary key) and "sport" (VARCHAR (50)). After the creation, it should appear as follows:



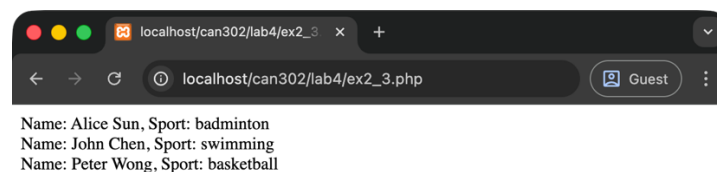
(c) Write a PHP script named "ex2_2.php" to insert three records as follows:

name	sport
Alice Sun	badminton
Peter Wong	basketball
John Chen	swimming

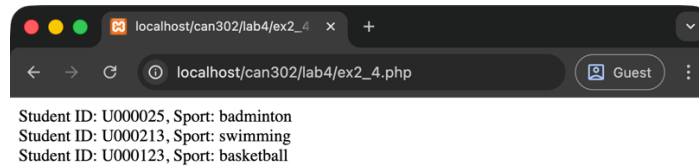
(d) After the insertion, it should appear as follows:



(e) Write a PHP script named "ex2_3.php" to query all records from the "hobby" table, and display their names and sports. It should appear as follows:

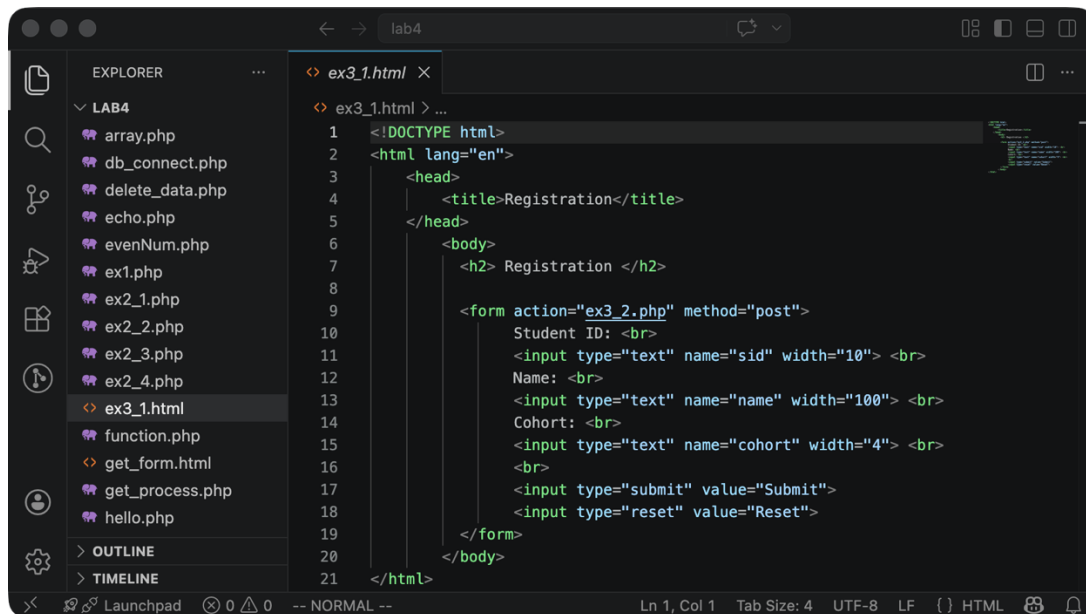


- (f) Write a PHP script named "ex2_4.php" to join the tables of "students" and "hobby", and display the student IDs along with their sports. It should appear as follows:

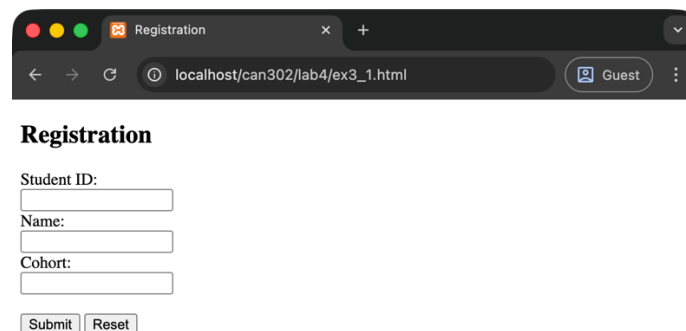


3. In this task, you are going to create a registration page with form and database.

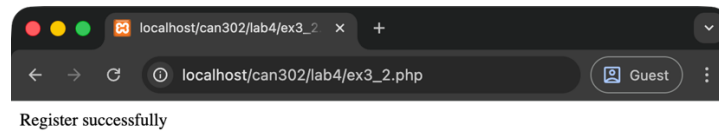
- (a) Create a new file named "ex3_1.html" as follows:



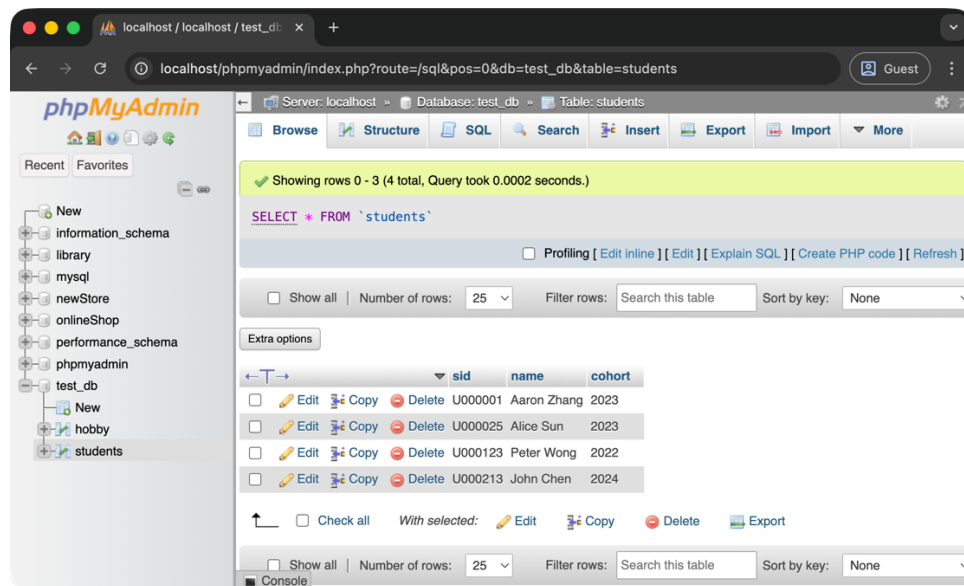
- (b) Open the browser and visit http://localhost/can302/lab4/ex3_1.html



- (c) Create a new file "ex3_2.php" to handle the form data and insert into the table "students" under "test_db".
- (d) Input a student ID, name, and cohort (e.g., U000001, Aaron Zhang, 2023), and click the "Submit" button:



- (e) Visit <http://localhost/phpmyadmin>, and check if the record is inserted:



Summary

We have achieved the following things in this lab session:

- Write server-side script programs in PHP
- Generate web pages content with functions
- Create tables in MySQL
- Access database in PHP